Course: MMDS

Lab 03

MapReduce

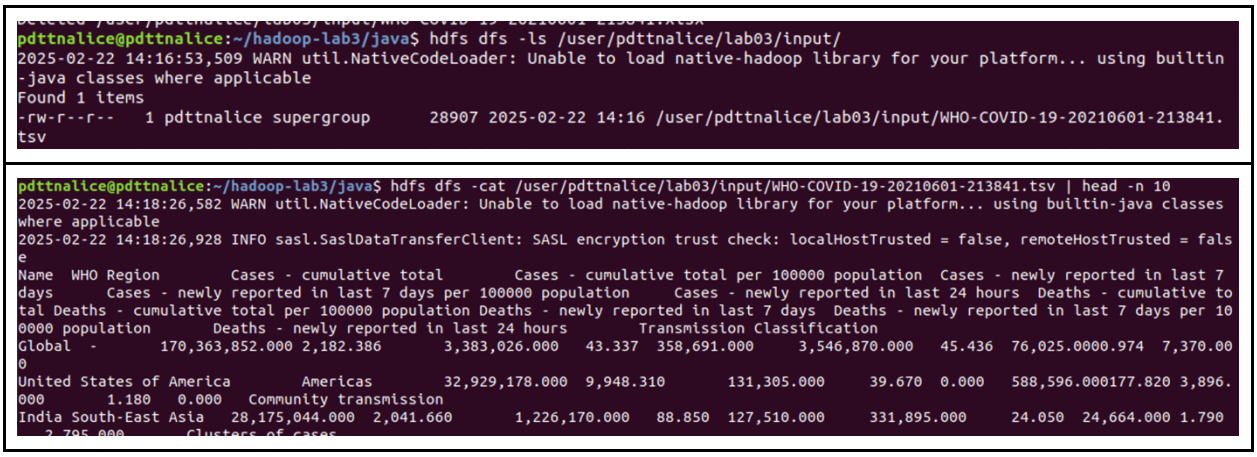
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Question 1:

Given a tsv file WHO-COVID-19-20210601-213841.tsv which is corresponding to the WHO Coronavirus (COVID-19) Dashboard.

Students are required to create a folder, named **lab03**, in HDFS and then copy the tsv to **lab03/input/**

Take a screenshot to show the content of **lab03/input/** in HDFS



Question 2:

Create one and only one java file, named **ASEANCaseCount.java**, to run a MapReduce job that counts the number of cumulative total cases among ASEAN countries (*South-East Asia Region in the given data table*).
The output of the MapReduce job is located in **lab03/output-java/**.
Submit the source code file following the instructions in Submission Notice.

```
pdttnalice@pdttnalice:~/hadoop-lab3/java$ hdfs dfs -cat /user/pdttnalice/lab03/output-java/part-r-00000
2025-02-22 14:56:41,077 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where
applicable
2025-02-22 14:56:41,443 INFO sasl.SaslDataTransferClient: SASL encryption trust check: localHostTrusted = false, remoteHostTrusted = false
Brunei Darussalam       243
Cambodia        30094
Indonesia       1821703
Lao People's Democratic Republic        1912
Malaysia        572357
Myanmar 143629
Philippines     1230301
Singapore       62051
Thailand        162022
Viet Nam        7432
pdttnalice@pdttnalice:~/hadoop-lab3/java$
```

Question 3 (*optional*):

Create a pair of Python files, named **ASEANDeathCountMapper.py** and **ASEANDeathCountReducer.py**, to run a MapReduce job that counts the number of cumulative total deaths among ASEAN countries (*South-East Asia Region in the given data table*).
The output of the MapReduce job is located in **lab03/output-python/**.
Submit the source code files following the instructions in Submission Notice.

```
pdttnalice@pdttnalice:~/hadoop-lab3/python$ hdfs dfs -cat /user/pdttnalice/lab03/output-python/part-00000
2025-02-22 16:32:33,354 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where
applicable
2025-02-22 16:32:33,692 INFO sasl.SaslDataTransferClient: SASL encryption trust check: localHostTrusted = false, remoteHostTrusted = false
Brunei Darussalam       3
Cambodia        214
Indonesia       50578
Lao People's Democratic Republic        3
Malaysia        2796
Myanmar 3217
Philippines     20966
Thailand        1069
Viet Nam        47
pdttnalice@pdttnalice:~/hadoop-lab3/python$
```